

# Ganesh Gupta

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## EXPERIENCE

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### Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Owned backend services and REST APIs (Python/FastAPI) handling millions of daily transactions at 99%+ uptime, with CI/CD automation and design reviews setting the bar for testability.
- Engineered a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation under scale constraints.
- Built a deployment-gated evaluation framework enforcing quality and regression checks across production outputs, blocking 15% of failed releases before traffic cutover.

### QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built an observability framework (Prometheus-style metrics) tracking drift, latency, and regressions in production, enabling governance-aligned rollback workflows.
- Cut p99 serving latency 420ms to 140ms across 50K+ daily requests by migrating synchronous services to an async, containerized architecture with autoscaling.

### Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected 25+ production Java Spring Boot microservices for financial investment workflows serving 1M+ users at 99.9% uptime over 4 years on AWS; led architecture decisions and code-review sessions.
- Architected and operated 25+ Java Spring Boot microservices on AWS (EC2, RDS, SQS, Lambda); led cloud modernization assessments and designed migration paths from legacy workloads toward a target-state architecture.
- Standardized REST API failure-handling contracts across 25+ Java microservices, reducing escalations 60% via consistent retry, timeout, and OAuth-aligned authentication patterns.
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.

## PROJECTS

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### Distributed KV Store / VPC Control Plane Simulator

[github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing and resource placement.

### Watch-your-LLM

[github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Full-stack platform with FastAPI backend, React/TypeScript dashboard, and PostgreSQL, with automated test coverage and CI/CD pipelines for release readiness.

### RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- Multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports, reducing manual debugging toil.

## EDUCATION

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University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

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**Languages:** Java, Python, TypeScript, C++, SQL

**Backend & Architecture:** Spring Boot, Microservices, REST APIs, Service Decomposition, OAuth, Redis, Kafka

**Data:** PostgreSQL, SQL/Relational Databases, NoSQL

**Cloud & DevOps:** AWS (EC2, S3, RDS, SQS, Lambda), Docker, Kubernetes, CI/CD, Observability (Prometheus/Grafana)

## CERTIFICATIONS

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AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer